

Context-Aware Music Understanding Using Graph Neural Networks and BERT: A Multimodal Fusion Approach

Niloy Ahsan
Dept. of CSE
BRAC University
Dhaka, Bangladesh
niloy.ahsan@g.bracu.ac.bd

Chaity Rani Ghosh
Dept. of CSE
BRAC University
Dhaka, Bangladesh
chaity.rani.ghosh@g.bracu.ac.bd

Abstract—Music understanding requires the integration of multiple forms of information, including audio structure, semantic descriptions, genre, mood, and contextual tags. This project investigates a multimodal approach that combines Graph Neural Networks (GNNs) and Bidirectional Encoder Representations from Transformers (BERT) to model complementary structural and semantic information in music. The project is organized into four progressive tasks. First, a BERT-based multi-label classifier is developed for textual music-tag understanding using the MagnaTagATune dataset. Second, a Graph Attention Network (GAT) is applied to audio-derived music structure graphs constructed from the FMA-small dataset. Third, the graph representation is combined with BERT representations to investigate multimodal context understanding. Finally, a contrastive GNN–BERT dual encoder is trained using MusicCaps to align audio graphs with natural language music descriptions in a shared embedding space. The final stage is evaluated using cross-modal retrieval metrics, including R@1, R@5, and R@10, together with qualitative retrieval examples and zero-shot tag prediction. The results show the feasibility of combining graph-based audio representations with transformer-based textual representations for music context understanding and cross-modal retrieval.

Index Terms—Music Understanding, Graph Neural Networks, BERT, Multimodal Learning, Music Graphs, Contrastive Learning, MusicCaps

I. INTRODUCTION

Music is a complex form of information that integrates multiple dimensions including acoustic characteristics, musical structure, genre, mood, instrumentation, and natural language descriptions. Understanding this information requires models that can capture both relationships within audio signals and the semantic information expressed through text. Conventional audio-based approaches, such as convolutional Neural Networks (CNNs) operating on spectrograms are effective at learning local acoustic patterns but do not explicitly represent relationships between different musical segments. Similarly, language models can capture semantic relationships in textual descriptions but do not directly model the structural properties of audio.

This project examines a multimodal approach for music context understanding by combining Graph Neural Networks (GNNs) with Bidirectional Encoder Representations from Transformers (BERT). The GNN component represents music

as a graph constructed from audio segments and learns structural representations through message passing, while BERT provides contextual representations from textual music information. The overall objective is to investigate how these complementary representations can be used for music tagging, context understanding, and cross-modal alignment.

The project is organized as a sequence of four tasks with increasing levels of complexity. Task 1 establishes a BERT-based baseline for multi-label music tag prediction using the MagnaTagATune dataset. Task 2 investigates audio-only graph learning using a Graph Attention Network (GAT) on music structure graphs constructed from the FMA-small dataset, with a CNN-based mel-spectrogram model used as a baseline. Task 3 combines the learned audio-graph and textual representations through GNN–BERT fusion for multimodal context understanding. Finally, Task 4 extends the framework to contrastive cross-modal learning using MusicCaps, where a dual-encoder architecture learns a shared embedding space between music graphs and natural language captions.

Through these progressive tasks, the project evaluates the effectiveness of graph-based audio representations, transformer-based semantic representations, multimodal fusion, and contrastive cross-modal learning. The experimental analysis includes classification metrics, baseline and ablation comparisons, embedding visualization, qualitative case studies, and cross-modal retrieval evaluation.

II. BACKGROUND AND PROJECT PIPELINE

The proposed project explores music understanding from two complementary perspectives: structural information extracted from audio and semantic information expressed through text. The structural component represents a music clip as a graph where nodes correspond to short audio segments, and edges represent relationships between segments. A Graph Neural Network (GNN) then propagates information between connected nodes to produce a graph-level representation. In this project, Graph Attention Networks (GATs) are used to learn different importance weights for neighboring nodes during message passing [1].

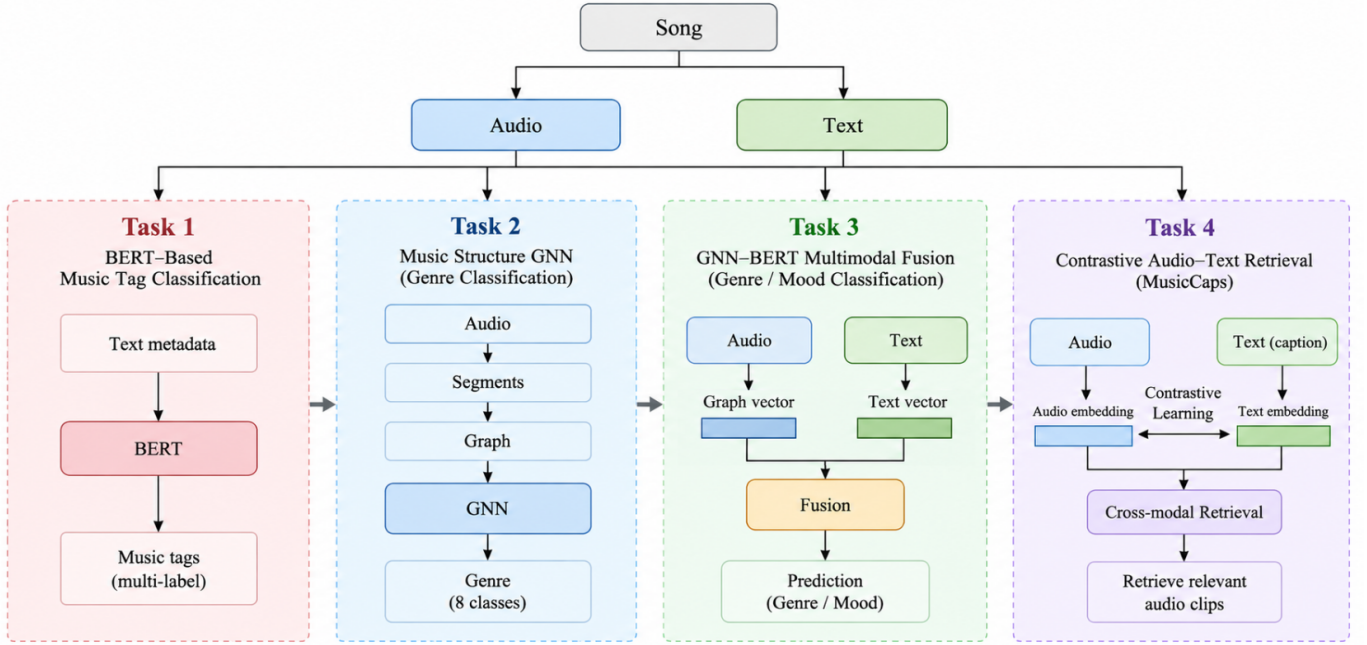


Fig. 1. Overview of the four-task multimodal music-understanding pipeline.

The semantic component is based on Bidirectional Encoder Representations from Transformers (BERT). BERT uses bidirectional transformer-based contextual representations to encode textual information [2] while its underlying transformer architecture relies on self-attention to model relationships between elements of a sequence [3]. In this project, BERT is used to represent textual music information, including metadata derived text and natural language MusicCaps descriptions.

The project is organized into four progressively advanced tasks. Task 1 establishes a textual baseline by fine-tuning BERT for multi-label music-tag prediction using MagnaTagATune. Task 2 introduces graph-based audio representation learning by constructing segment-level music graphs from FMA-small audio and training a GAT-based classifier, with a CNN operating on mel-spectrograms as an audio baseline. Task 3 combines graph and textual representations through GNN-BERT fusion for multimodal music-context understanding. Task 4 extends the framework to cross-modal representation learning by training a GNN-BERT dual encoder with contrastive learning on paired MusicCaps audio clips and captions.

The complete progression of the proposed framework is illustrated in Figure 1. It shows how the project moves from independent textual and audio representations in Tasks 1 and 2 to supervised multimodal fusion in Task 3 and contrastive audio-text alignment in Task 4.

III. DATASET DESCRIPTION AND PREPROCESSING

The project uses three principal datasets across the four tasks: MagnaTagATune for textual music-tag understanding,

the Free Music Archive (FMA-small) for audio-based graph learning and supervised multimodal classification, and MusicCaps for cross-modal audio-caption alignment. These datasets provide complementary forms of information, ranging from human-generated music tags to genre labels and natural language descriptions.

A. MagnaTagATune

MagnaTagATune is a music information retrieval dataset containing human-generated annotations collected through the TagATune game. The dataset provides binary annotations over a broad set of music-related tags, including genre, instrumentation, and perceptual characteristics [4]. The annotation file used in this project contains 25,863 annotated clips and 188 available tags.

For Task 1, tag frequencies were calculated from the annotation matrix and the 50 most frequent tags were selected as prediction targets. The resulting labels form a 50-dimensional multi-label target vector for each clip. Textual input was constructed from the corresponding metadata as

$$X_{\text{text}} = \text{title} + \text{" by " + artist + \text{" from " + album} \quad (1)$$

The resulting dataset was divided into training, validation, and test sets using two consecutive random splits with a fixed random seed of 42. The first split reserved 20% of the samples for testing while about 20% of the remaining samples were used for validation. So, this result produced 16,552 training samples, 4,138 validation samples, and 5,173 test samples.

The text was tokenized using the uncased BERT tokenizer with padding, truncation, and a maximum sequence length of 128 tokens in the project. The resulting token IDs were used as input to the BERT-based multi-label classifier. The main dataset statistics are summarized in Table I.

TABLE I
MAGNATAGATUNE STATISTICS AND TASK 1 SPLITS.

Property	Value
Annotated clips	25,863
Available tags	188
Selected tags	50
Training samples	16,552
Validation samples	4,138
Test samples	5,173

B. FMA-small

The Free Music Archive (FMA) is a publicly available music dataset which is designed for music information retrieval research [5]. Its small subset portion contains 8,000 30-second audio clips distributed equally among eight genres. The project uses all eight FMA-small genres such as Electronic, Experimental, Folk, Hip-Hop, Instrumental, International, Pop, and Rock.

The official FMA split information was used to construct the training, validation, and test partitions. The resulting dataset contains 6,400 training tracks, 800 validation tracks, and 800 test tracks. Audio files were loaded as mono signals at a sampling rate of 22,050 Hz and limited to 30 seconds.

In graph construction each audio track was divided into overlapping 3-second segments using a 1.5-second hop. Each segment was represented using a 138-dimensional feature vector. The features consist of the mean and standard deviation of 20 MFCC coefficients, MFCC delta coefficients, 12 chroma features, 6 tonnetz features, and 7 spectral-contrast features, together with mean and standard deviation statistics for spectral centroid, spectral rolloff, zero-crossing rate, and RMS energy. The complete feature composition is given in Table II.

TABLE II
COMPOSITION OF THE 138-DIMENSIONAL AUDIO FEATURES.

Feature	Dimensions
MFCC mean and standard deviation	40
MFCC delta mean and standard deviation	40
Chroma mean and standard deviation	24
Tonnetz mean and standard deviation	12
Spectral contrast mean and standard deviation	14
Spectral centroid mean and standard deviation	2
Spectral rolloff mean and standard deviation	2
Zero-crossing rate mean and standard deviation	2
RMS energy mean and standard deviation	2
Total	138

Here, each audio clip is represented as a graph,

$$G = (V, E) \quad (2)$$

where V denotes the set of audio segments and E contains temporal and acoustic-similarity connections. Consecutive temporal segments are connected bidirectionally with an

edge weight of ‘1.0’. In addition, each segment is connected bidirectionally to its three nearest neighbors according to cosine similarity between their feature vectors.

The node features were standardized using the mean and standard deviation calculated from the training graphs only.

$$x' = \frac{x - \mu_{\text{train}}}{\sigma_{\text{train}}} \quad (3)$$

The same training statistics were also applied to the validation and test graphs. The normalization process was retained when the Task 2 graph encoder was reused for the subsequent Task 3 experiments.

During graph construction, six FMA tracks could not be successfully processed by the audio loader. Consequently, the final graph cache contained 6,394 training graphs while the validation and test sets retained 800 graphs each. The official split sizes and successfully constructed graph counts are summarized in Table III.

TABLE III
FMA-SMALL SPLITS, GENRES, AND GRAPH COUNTS.

Split	Audio Tracks	Graphs
Training	6,400	6,394
Validation	800	800
Test	800	800
Total	8,000	7,994

C. MusicCaps

MusicCaps provides natural language descriptions of short music clips. The dataset consists of 5,521 examples with each example associated with a music caption and metadata describing the corresponding audio clip [6]. The examples correspond to approximately 10 second music excerpts.

The project loaded the MusicCaps dataset through the Hugging Face dataset interface. Because the experiments required locally available audio corresponding to the metadata only examples for which the required WAV clip had been successfully extracted were retained. After filtering for available audio, 951 paired audio caption examples remained.

The retained examples were randomly shuffled using a fixed random seed of 42 and divided into 80% training, 10% validation, and 10% test partitions. This resulted in 760 training pairs, 95 validation pairs, and 96 test pairs.

For audio preprocessing, the clips were converted to mono and resampled to 22,050 Hz. The same 3-second segment length and 1.5-second hop used for the FMA graph representation were applied to the MusicCaps audio. Each segment was represented using the same 138-dimensional feature vector, and temporal and cosine similarity edges were constructed in the same way as the FMA graphs.

The resulting MusicCaps graph collections contained 760 training graphs, 95 validation graphs, and 96 test graphs. The captions were tokenized using the BERT tokenizer with truncation and a maximum sequence length of 128 tokens. The resulting dataset statistics are summarized in Table IV.

TABLE IV
MUSICCAPS FILTERING, SPLITS, AND GRAPH STATISTICS.

Property	Value
Original examples	5,521
Examples with available audio	951
Training pairs	760
Validation pairs	95
Test pairs	96
Graph node feature dimension	138

D. Summary of the Preprocessing Pipeline

The preprocessing stages establish a consistent representation across the progressively more advanced tasks. MagnaTagATune provides metadata-derived text and multi-label targets for the BERT baseline. FMA-small provides labeled audio for graph-based genre classification and multimodal fusion. MusicCaps provides paired audio and natural language descriptions for contrastive cross-modal learning. The FMA and MusicCaps graph construction procedures use the same 138 dimensional segment representation allowing the graph encoder developed for supervised audio classification to be extended to the cross-modal setting.

IV. METHODOLOGY

The proposed framework consists of four progressive tasks that move from unimodal music understanding to multimodal representation learning. The same pretrained components are reused across tasks where appropriate.

A. Task 1: BERT-Based Music Tag Classification

Task 1 establishes the textual baseline using bert-base-uncased. For each music clip, the title, artist, and album metadata are combined into a single text sequence and tokenized with a maximum length of 128 tokens. BERT produces a 768-dimensional [CLS] representation which is passed to a 50 label classification layer. Because multiple tags can be associated with a single clip, sigmoid activation and binary cross-entropy loss are used for multi-label prediction. The 50 target tags are selected according to their frequency in the MagnaTagATune annotations. The resulting fine tuned BERT encoder is reused in Tasks 3 and 4.

B. Task 2: Audio Music Graph and GNN

Task 2 represents each FMA-small audio clip as a graph $G = (V, E)$ where each node corresponds to an overlapping 3 second audio segment with a 1.5 second hop. Each segment is represented by a 138 dimensional acoustic feature vector.

Two types of edges are used: bidirectional temporal edges between consecutive segments and bidirectional similarity edges connecting the three nearest neighbors according to cosine similarity. Node features are standardized using statistics calculated from the training set.

A two layer GATv2 encoder processes the resulting graphs. The first layer uses 48 channels with two attention heads, followed by normalization, ReLU, and dropout. A second

graph-attention layer produces 48 dimensional node representations. Global mean and maximum pooling are concatenated to obtain a 96 dimensional graph representation which is passed to an eight-class genre classifier. A CNN operating on mel-spectrograms is used as the audio only baseline for comparison. The audio-graph construction process is illustrated in Figure 2.

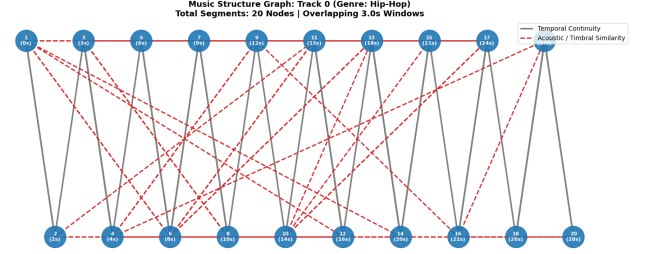


Fig. 2. Audio-graph construction using temporal and similarity edges.

C. Task 3: GNN-BERT Multimodal Fusion

Task 3 combines the structural representation from the GNN with the semantic representation from BERT. The graph encoder produces a 96-dimensional representation, while the fine-tuned BERT encoder produces a 768-dimensional representation from the corresponding FMA metadata.

Two fusion strategies are evaluated. In early concatenation, both representations are projected to 256 dimensions and concatenated before classification. In cross-attention, the projected graph representation is used as a query over the token-level BERT representations allowing the graph representation to attend selectively to textual information. The experiments compare GNN-only, BERT-only, early-concatenation, and cross-attention configurations. All models perform eight-class FMA-small genre classification and are evaluated using Accuracy, Macro-F1, and AUC-PR.

D. Task 4: Contrastive Audio-Text Learning

Task 4 learns a shared representation space between music graphs and natural language descriptions using paired MusicCaps examples. The GNN produces a 96-dimensional graph embedding, while the fine-tuned BERT encoder produces a 768-dimensional caption embedding. Separate projection heads map both representations into a shared 256-dimensional space. The embeddings are normalized and compared using temperature-scaled cosine similarity. A symmetric InfoNCE loss is used to optimize both graph-to-text and text-to-graph retrieval:

$$\mathcal{L} = \frac{1}{2} [\text{CE}(S, I) + \text{CE}(S^T, I)], \quad (4)$$

where S is the graph-text similarity matrix and the temperature is set to $\tau = 0.07$.

During training, the final two BERT encoder layers are fine tuned with a learning rate of 10^{-6} , while the GNN and projection heads use a learning rate of 5×10^{-5} . Early

stopping is based on validation loss. The learned shared space is evaluated through cross-modal retrieval using R@1, R@5, and R@10. Qualitative retrieval examples are also examined to assess the semantic correspondence between captions and retrieved music clips.

E. Implementation Summary

All experiments were implemented using PyTorch, PyTorch Geometric, and the Hugging Face Transformers library. A fixed random seed of 42 was used for dataset splitting where applicable. The graph construction and acoustic feature extraction pipeline was kept consistent across Tasks 2, 3, and 4 while the fine tuned BERT encoder from Task 1 was reused for the subsequent multimodal experiments.

V. RESULTS ANALYSIS

A. Task 1: BERT-Based Music Tag Classification

The fine-tuned BERT model was evaluated on the held-out MagnaTagATune test set using element-wise accuracy, Macro-F1, Micro-F1, and mean AUC-PR. The model achieved an accuracy of 0.8909, Macro-F1 of 0.3629, Micro-F1 of 0.4048, and mean AUC-PR of 0.3341. The final test results are summarized in Table V.

TABLE V
TASK 1 PERFORMANCE ON THE MAGNATAGATUNE TEST SET.

Metric	Score
Accuracy	0.8909
Macro-F1	0.3629
Micro-F1	0.4048
Mean AUC-PR	0.3341

The training behavior of the fine-tuned BERT model is shown in Figures 3, 4, and 5. The training and validation loss curves show the optimization behavior, while the F1 and AUC-PR curves illustrate the evolution of multi-label classification performance across epochs.



Fig. 3. Training and validation loss of the Task 1 BERT model across training epochs.

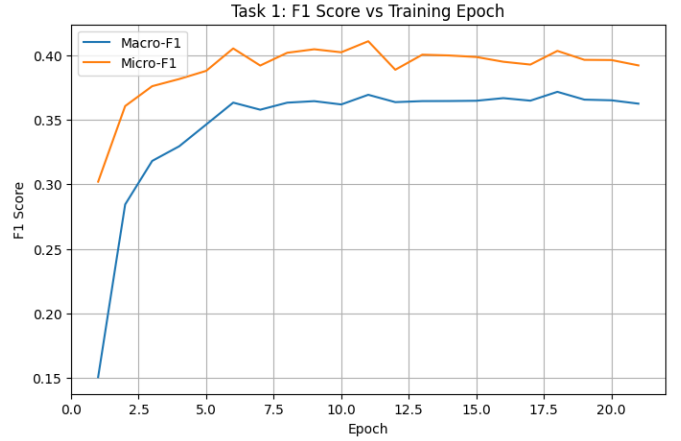


Fig. 4. Macro-F1 and Micro-F1 scores of the Task 1 BERT model across training epochs.

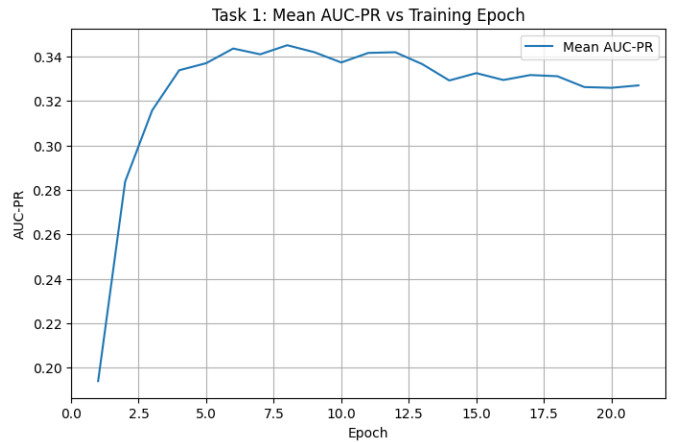


Fig. 5. Mean AUC-PR of the Task 1 BERT model across training epochs.

The high element-wise accuracy should be interpreted together with the F1-based metrics because the task is multi-label and contains substantially more negative than positive labels. Macro-F1, Micro-F1, and AUC-PR therefore provide a more informative view of the model's ability to identify relevant music tags.

B. Task 2: Music Structure GNN and CNN Baseline

The GNN was evaluated on the eight FMA-small genre classes and compared with a CNN operating on mel-spectrograms. As shown in Table VI, the GNN achieved a test accuracy of 0.4462 and Macro-F1 of 0.4416, while the CNN achieved 0.5500 accuracy and 0.5362 Macro-F1.

TABLE VI
TASK 2 GNN AND CNN TEST PERFORMANCE.

Model	Accuracy	Macro-F1
Music Structure GNN	0.4462	0.4416
2D-CNN Mel-Spectrogram	0.5500	0.5362

The confusion matrix of the GNN and CNN predictions are compared in Figure 6. This visualization shows the class-level prediction patterns of the two audio models.

The training behavior and precision-recall performance of the GNN are shown in Figure 11. The figure presents training and validation loss, validation and test accuracy, and the class-wise precision-recall curves.

The CNN outperformed the music structure GNN on both evaluation metrics. This indicates that under the feature extraction and graph construction strategy used in this experiment, the handcrafted segment level graph representation did not preserve as much discriminative information for genre classification as the mel-spectrogram representation. The result also provides a useful baseline for evaluating whether graph-based structural modeling offers an advantage over conventional time-frequency representations.

C. Task 3: GNN-BERT Multimodal Fusion

Task 3 evaluates GNN-only, BERT-only, early concatenation, and cross-attention configurations. The results are summarized in Table VII. Early concatenation achieved the strongest overall performance with an accuracy of 0.4625, Macro-F1 of 0.4613, and AUC-PR of 0.4922.

TABLE VII
TASK 3 ABLATION RESULTS ON FMA-SMALL.

Model	Accuracy	Macro-F1	AUC-PR
GNN-only	0.4363	0.4334	0.4488
BERT-only	0.2812	0.2752	0.3010
Early Concat	0.4625	0.4613	0.4922
Cross-Attention	0.4363	0.4335	0.4644

The confusion matrix and precision-recall curves for the early concatenation model are shown in Figures 7 and 8 respectively. These visualizations provide class-level and precision-recall analysis for the best-performing fusion configuration.

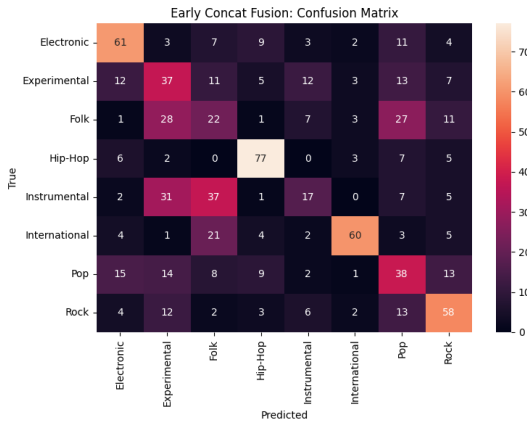


Fig. 7. Early Concat confusion matrix.

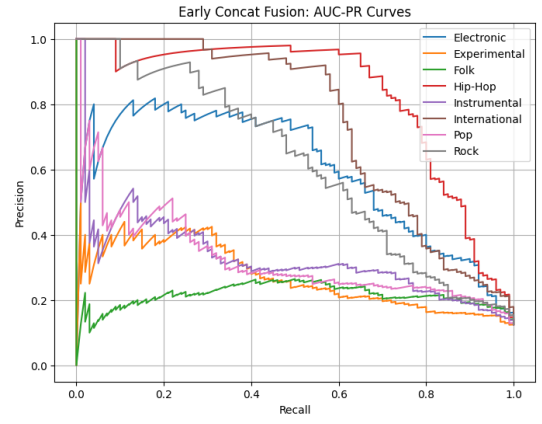


Fig. 8. Early Concat PR curve.

For comparison, the cross-attention model also produced a confusion matrix and class-wise precision-recall curves. These results are shown in Figures 9 and 10. Although cross-attention achieved a slightly higher AUC-PR than GNN-only, it did not improve accuracy or Macro-F1 over the simpler early-concatenation strategy.

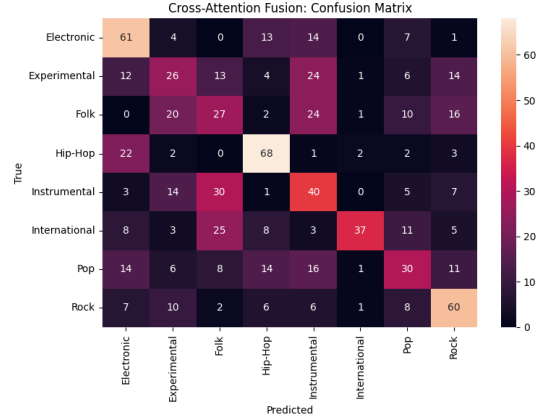


Fig. 9. Cross-Attention confusion matrix.

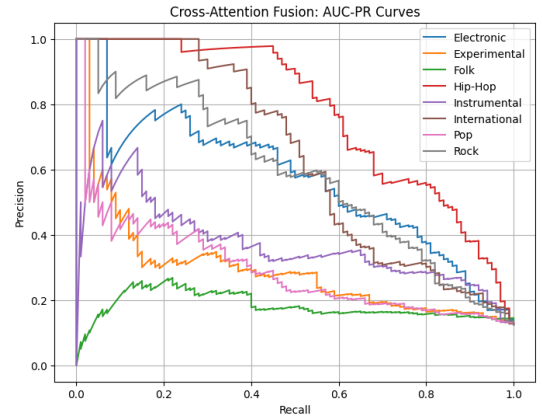


Fig. 10. Cross-Attention PR curve.

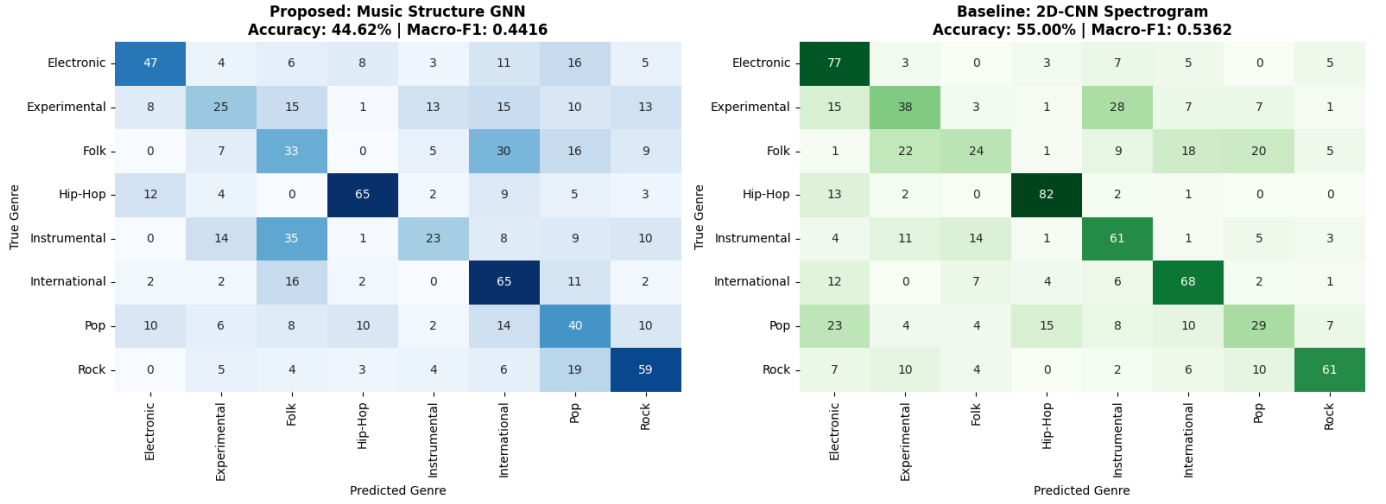


Fig. 6. Confusion matrix of the GNN and CNN predictions.

Early concatenation improved Macro-F1 from 0.4334 to 0.4613 compared with the GNN-only model and achieved the highest AUC-PR among the evaluated configurations. In contrast, the BERT-only model obtained the lowest performance, suggesting that the metadata-derived textual representation alone was insufficient for the FMA-small genre classification task. Cross-attention achieved a slightly higher AUC-PR than GNN-only but did not improve accuracy or Macro-F1 and did not outperform early concatenation. Thus, in this experiment, simple representation-level fusion was more effective than the implemented cross-attention mechanism.

1) *Task 3 Representation Visualization*: The learned fusion representations were visualized using t-SNE to examine the organization of the multimodal representations by genre and mood. The genre-based visualization in Figure 12 provides a qualitative view of how samples from different FMA-small genres are distributed in the learned representation space.

The mood-based representation is shown in Figure 13. However, 767 of the 800 test samples were assigned an unknown mood label. Therefore, this visualization provides limited evidence for assessing mood-level separation and should be interpreted cautiously.

2) *Task 3 Case Studies*: Three test examples were examined qualitatively to relate the learned representations to their corresponding audio graphs and textual metadata. The selected examples represent the International, Experimental, and Folk genres. Their corresponding audio-graph structures are presented together in Figure 14.

The graphs contain 20 audio-segment nodes for each selected track. The visualizations illustrate the temporal and similarity relationships present in the constructed music graphs. The displayed paths represent temporal ordering rather than learned attention importance.

D. Task 4: Contrastive Audio-Text Retrieval

The contrastive GNN-BERT model was evaluated on the held-out MusicCaps test set using cross-modal retrieval. As shown in Table VIII, the model achieved R@1 of 5.21%, R@5 of 17.71%, and R@10 of 31.25%.

TABLE VIII
TASK 4 RETRIEVAL PERFORMANCE ON MUSICCAPS.

Metric	Score
R@1	5.21%
R@5	17.71%
R@10	31.25%

The increasing R from R@1 to R@10 indicates that the shared embedding space can identify relevant audio clips within a larger retrieval candidate set. However, the relatively low R@1 shows that precise audio-caption matching remains challenging. This is particularly relevant for MusicCaps because captions can describe several musical properties simultaneously, including instrumentation, genre, vocal characteristics, mood, and production style.

1) *Qualitative Retrieval Analysis*: Ten qualitative retrieval examples were examined by comparing each MusicCaps caption with the highest-ranked retrieved audio clips. The results are summarized in Table IX. The examples show varying degrees of semantic correspondence. Some queries produced closely related musical descriptions, while others returned clips with only partial semantic overlap.

The qualitative examples show that the learned representation captures some broad semantic relationships between captions and audio graphs but retrieval errors remain substantial. Exact or highly similar matches appear in a small number of examples whereas other queries retrieve clips that share only individual musical characteristics. These observations are consistent with the quantitative retrieval results and indicate that the contrastive representation provides a meaningful but

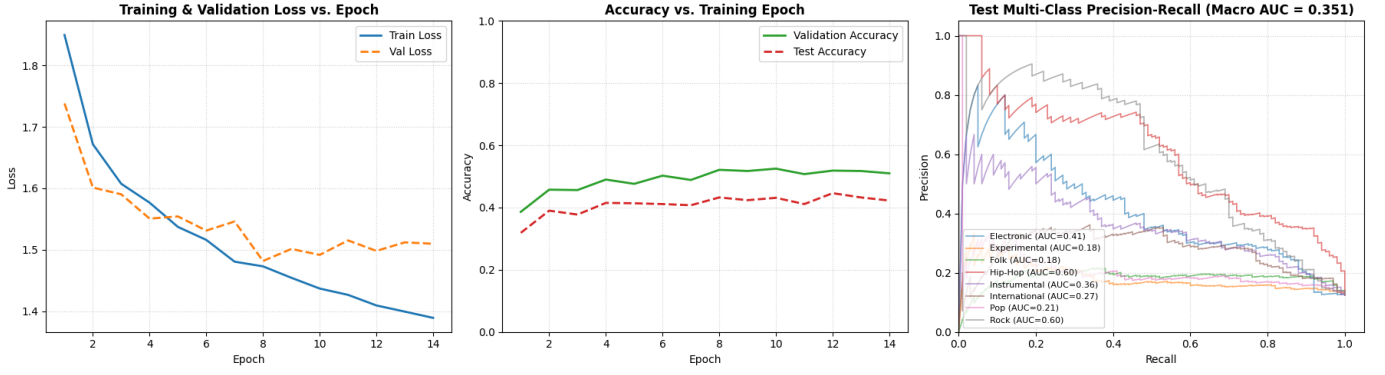


Fig. 11. Training and precision-recall performance of the GNN for Task 2, showing (a) training and validation loss, (b) validation and test accuracy, and (c) class-wise precision-recall curves.

TABLE IX
MUSICCAPS CAPTION-TO-AUDIO RETRIEVAL EXAMPLES.

Query Caption	Top Retrieved Descriptions
Arabic folk music	Country music; heavy metal; pop music
Sad rock with theremin	Marimba music; power metal; rap music
Accordion with a groovy character	Pop music; slow romantic acoustic guitar; marimba music
Relaxed female vocal with sensual character and scratching	Drone music; accordion and groovy music; mellow rap
Hip-hop with a crime/noir character	Anime music; energetic instrumental; sensual speech/scratching
Horror sound effects	Anime music; sinister electronic music; synth/electronic music
Digital drums, sub-bass, rapping, and marimba	Electro pop; processed vocal/synth music; matching caption
Resonating bowl	Matching caption; shofar; theremin
Power metal	Sinister electronic music; rap with strings; electronic horror music
Gothic rock	Electronic horror music; rap with strings; matching caption

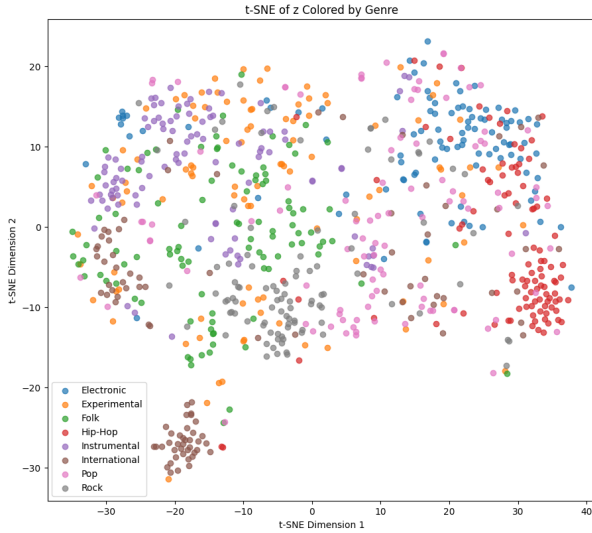


Fig. 12. Task 3 t-SNE visualization by genre.

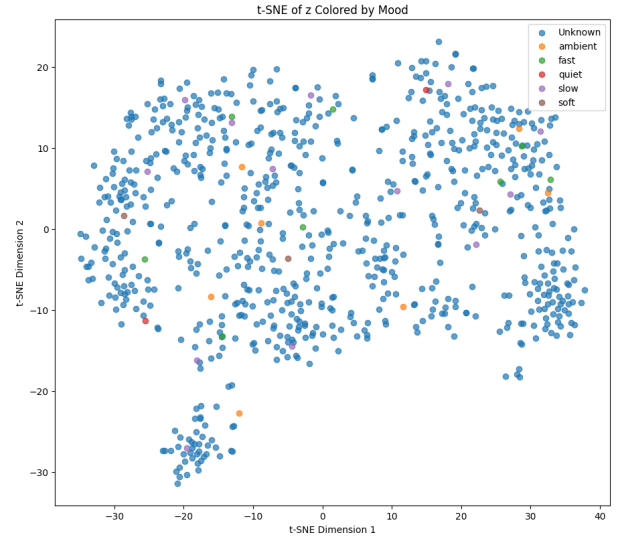


Fig. 13. Task 3 t-SNE visualization by mood label.

still imperfect alignment between audio structure and natural language descriptions.

E. Overall Analysis

Across the four tasks, the experiments show a progression from unimodal music understanding toward multimodal representation learning. Task 1 demonstrates that BERT can extract

useful information from metadata-derived text for multi-label music-tag prediction. Task 2 shows that the proposed graph representation can perform genre classification, although the mel-spectrogram CNN provides a stronger audio-only baseline. Task 3 demonstrates that combining graph and textual representations through early concatenation improves over

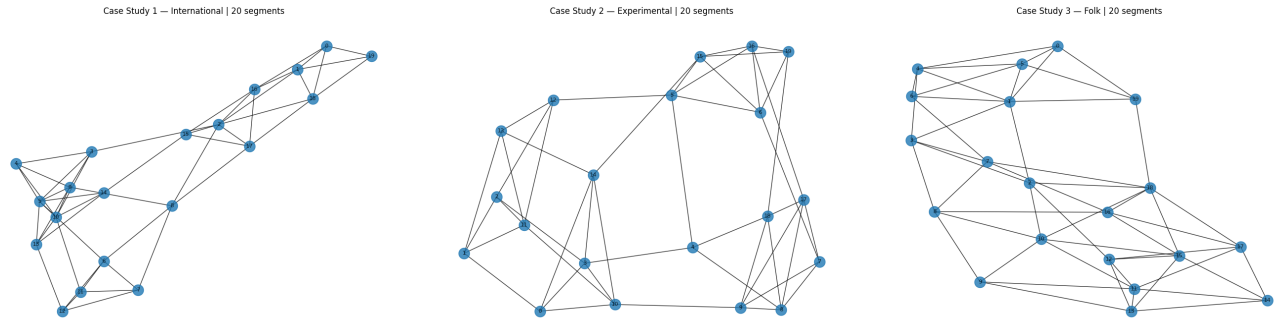


Fig. 14. Task 3 graph examples: (a) International, (b) Experimental, and (c) Folk.

the GNN-only configuration, while the implemented cross-attention model does not provide an additional performance advantage. Finally, Task 4 demonstrates the feasibility of learning a shared audio–text representation space for MusicCaps retrieval, although the relatively low R@1 indicates that more effective cross-modal alignment remains necessary.

VI. DISCUSSION

The experimental results show that the four tasks provide complementary perspectives on music context understanding. Task 1 demonstrates that textual metadata contains useful information for multi-label music-tag prediction, although the moderate Macro-F1 and AUC-PR indicate that predicting multiple music characteristics from metadata remains challenging.

For Task 2, the mel-spectrogram CNN outperformed the music structure GNN in both accuracy and Macro-F1. This suggests that the handcrafted segment-level graph representation may not preserve all of the discriminative acoustic information available in the spectrogram representation. Nevertheless, the GNN provides an explicit structural representation of temporal and acoustic similarity relationships between music segments, providing a basis for the subsequent multimodal experiments.

The Task 3 ablation demonstrates that combining graph and textual representations can improve classification performance when an appropriate fusion strategy is used. Early concatenation achieved the best performance among the evaluated configurations, outperforming both the GNN-only and BERT-only models. In contrast, the cross-attention model did not outperform early concatenation and achieved performance close to the GNN-only configuration. This result indicates that the simpler fusion strategy was more effective for the available data and learned representations in the current experiment.

Task 4 demonstrates the feasibility of learning a shared audio text representation through contrastive learning. The retrieval results show that the learned embedding space can associate natural language descriptions with relevant audio clips, with R@10 reaching 31.25%. However, the relatively low R@1 of 5.21% indicates that precise audio–caption alignment remains challenging. The qualitative retrieval examples further show that the model can capture some broad musical

characteristics, although exact semantic matching is inconsistent.

Overall, the experiments highlight the complementary roles of acoustic structure and textual semantics in music understanding. They also show that increasing architectural complexity does not necessarily guarantee better performance: in the present setting, early multimodal concatenation performed better than the more complex cross-attention strategy. The results therefore support a progressive approach in which graph-based acoustic representations and pretrained language representations are first evaluated independently and then combined according to the requirements of the target task.

VII. LIMITATIONS

Despite, the promising results, several limitations should be considered. First, the graph representation relies on handcrafted features from fixed-length audio segments, which may not capture higher-level harmonic, rhythmic, or structural relationships. Second, the graph-based model performed below the mel-spectrogram CNN in Task 2, suggesting that the current graph representation may not preserve all discriminative acoustic information available in spectrograms. Third, Task 4 was limited by the availability of paired audio–text data. Only 951 of the 5,521 MusicCaps examples had locally available audio, reducing the amount of data available for contrastive training. Finally, cross-attention did not outperform early concatenation, while sparse mood annotations limited the interpretation of the Task 3 mood visualization. These limitations suggest that larger paired datasets, stronger pretrained audio representations, richer annotations, and improved fusion strategies could further improve the framework.

VIII. CONCLUSION AND FUTURE WORK

This project presented a progressive framework for music context understanding using textual modeling, graph-based audio representation, multimodal fusion, and contrastive audio text learning. Task 1 showed that BERT can extract useful information from music metadata for multi-label tag prediction. Task 2 demonstrated the feasibility of representing audio as segment-level graphs, although the mel-spectrogram CNN achieved stronger genre classification performance. Task 3 showed that combining GNN and BERT representations

through early concatenation produced the best multimodal classification performance among the evaluated configurations. Task 4 further demonstrated the feasibility of learning a shared representation space between music graphs and natural language descriptions, achieving measurable cross-modal retrieval performance on MusicCaps.

Future work will focus on learning richer audio representations instead of relying primarily on handcrafted acoustic feature as well as developing more expressive graph construction strategies that can capture long range musical structure. Larger paired audio–text datasets could improve contrastive representation learning and retrieval quality. Further work of multimodal fusion mechanisms including improved cross-attention designs may also strengthen interactions between structural and textual information. Finally, more complete mood annotations and improved interpretability methods could support deeper analysis of musical context and the learned representations.

REFERENCES

- [1] P. Veličković, G. Cucurull, A. Casanova, A. Romero, P. Liò, and Y. Bengio, “Graph attention networks,” in *International Conference on Learning Representations*, 2018.
- [2] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, “Bert: Pre-training of deep bidirectional transformers for language understanding,” in *Proceedings of NAACL-HLT*, 2019.
- [3] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, and I. Polosukhin, “Attention is all you need,” in *Advances in Neural Information Processing Systems*, 2017.
- [4] E. Law, K. West, M. Mandel, M. Bay, and J. S. Downie, “Evaluation of algorithms using games: The case of music annotation,” in *Proceedings of the 10th International Society for Music Information Retrieval Conference*, 2009.
- [5] M. Defferrard, K. Benzi, P. Vandergheynst, and X. Bresson, “Fma: A dataset for music analysis,” in *Proceedings of the 18th International Society for Music Information Retrieval Conference*, 2017.
- [6] A. Agostinelli, T. I. Denk, Z. Borsos, J. Engel, M. Verzett, A. Caillon, Q. Huang, A. Jansen, A. Roberts, M. Tagliasacchi *et al.*, “Musiclm: Generating music from text,” *arXiv preprint arXiv:2301.11325*, 2023.